

NanoBanana Study Report: polyp | stage0 | nanobanana_polyp_s

run_dir: D:\Projects\gemini_image_segmentation\results_nanobanana\polyp\gemini-3.1-flash-image-pr

run_id: nanobanana_polyp_smoke_bw_closed_stage0_20260302-013204

stage: stage0

dataset: polyp

target: polyp

model: gemini-3.1-flash-image-preview

n_tasks: 10

n_rows: 10

n_failures: 0

workers: 6

Interpretation notes:

- PRIMARY excludes duplicate/mask-source flagged retrieval runs per config.
- SENSITIVITY includes all audited rows.
- Delta tables compare each tool mode against CLOSED on paired rows.

Summary By Tool Mode (PRIMARY)

tool_mode	n	mean_iou	mean_dice	mean_precision	mean_recall	qc_pass_rate	iou_ci_lower	iou_ci_upper	dice_ci_lower	dice_ci_upper
closed	10.0000	0.7836	0.8610	0.9557	0.8187	0.0000	0.6448	0.9037	0.7595	0.9413

Summary By Tool Mode (SENSITIVITY)

tool_mode	n	mean_iou	mean_dice	mean_precision	mean_recall	qc_pass_rate	iou_ci_lower	iou_ci_upper	dice_ci_lower	dice_ci_upper
closed	10.0000	0.7836	0.8610	0.9557	0.8187	0.0000	0.6484	0.9004	0.7498	0.9474

Delta Vs CLOSED (PRIMARY)

No data available.

QC Failure Summary (PRIMARY)

tool_mode	qc_fail_count	total	qc_fail_rate
closed	10.0000	10.0000	1.0000

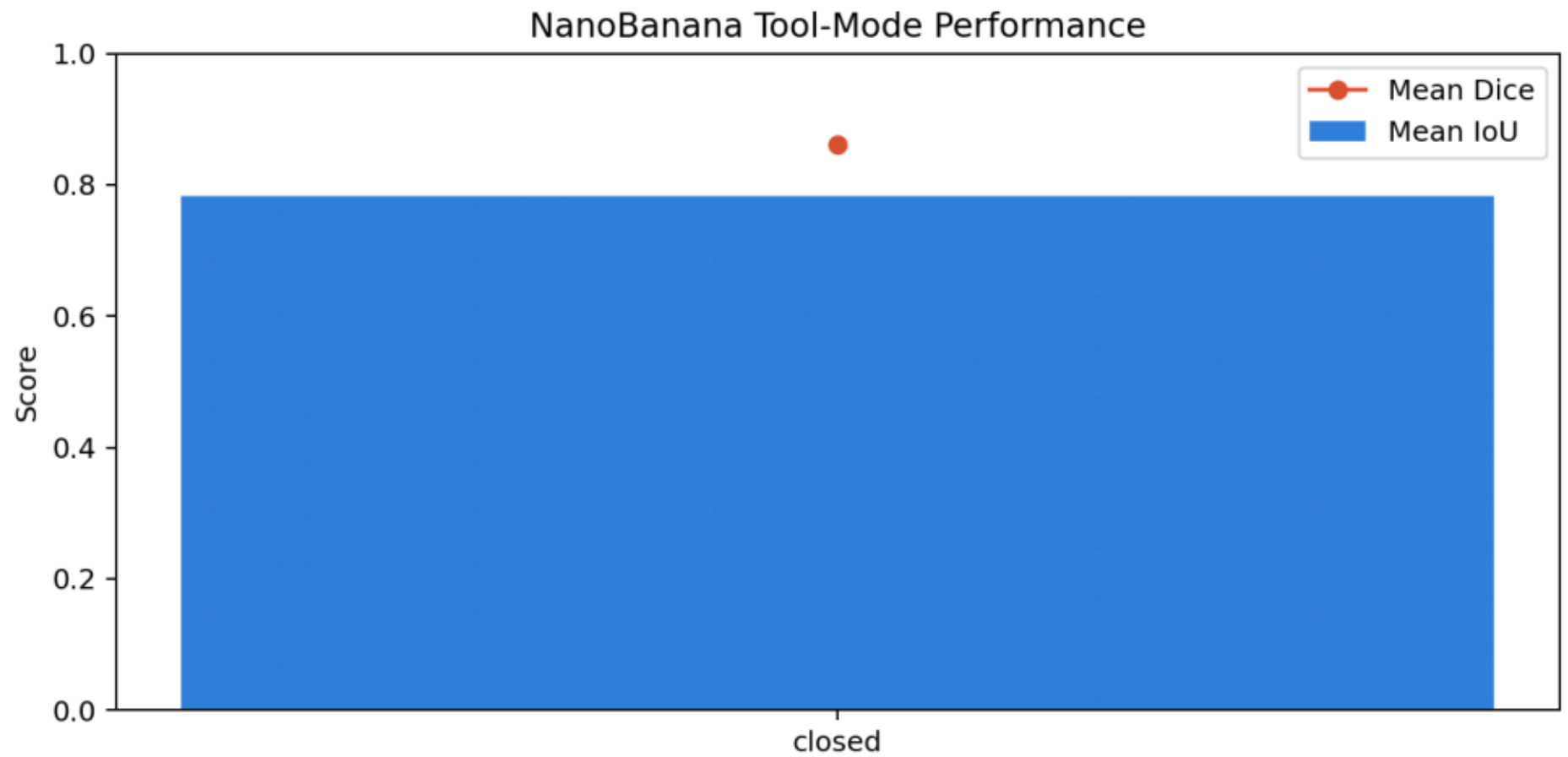
Retrieval Audit Summary

tool_mode	duplicate_rate	mask_source_rate	audit_unavailable_rate	audit_unavailable_count	n
closed	0.0000	0.0000	1.0000	10.0000	10.0000

Analysis Partition Counts

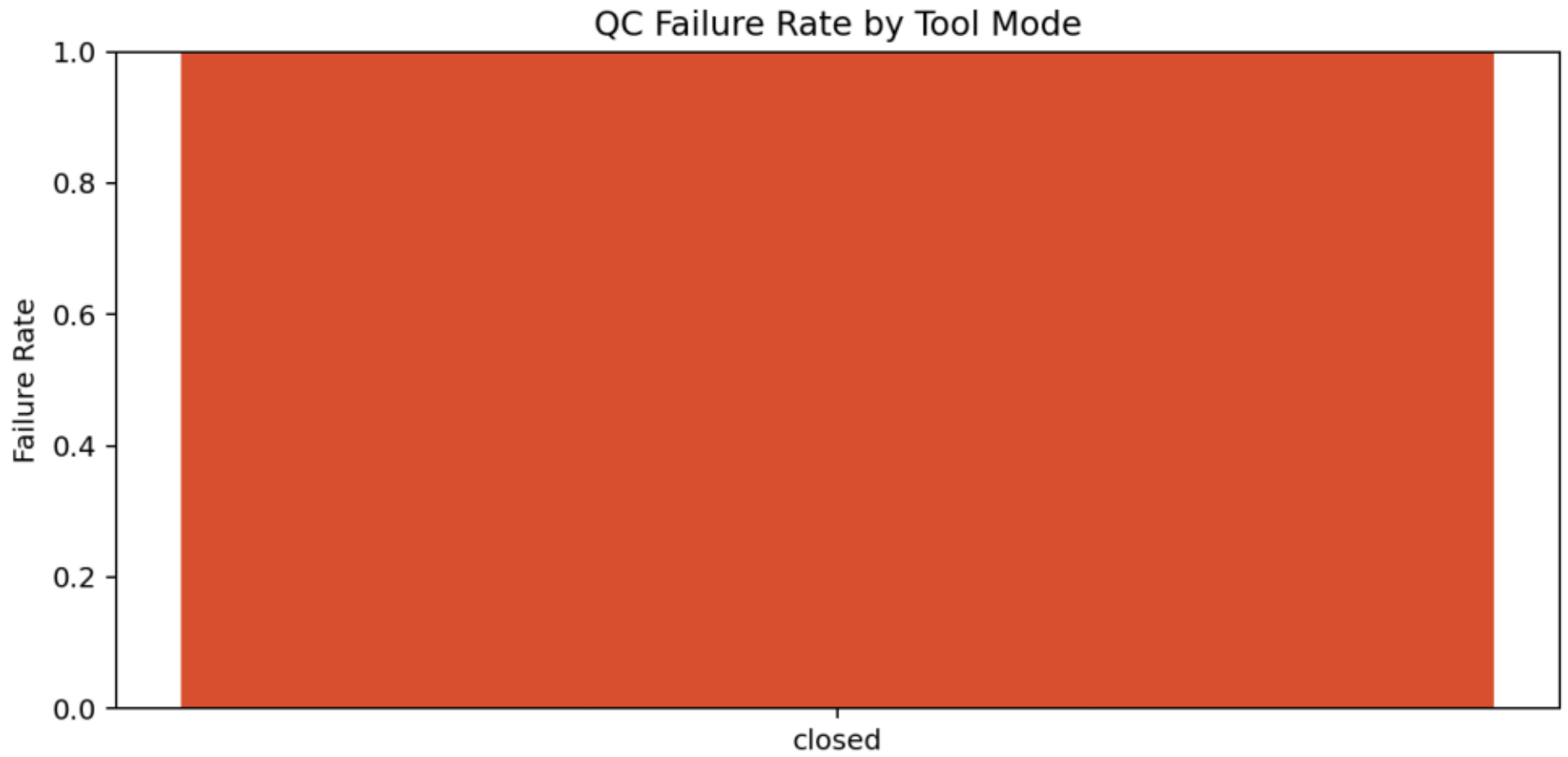
tool_mode	n_total	n_primary	n_sensitivity	n_audit_unavailable
closed	10.0000	10.0000	10.0000	10.0000

Primary Performance Plot

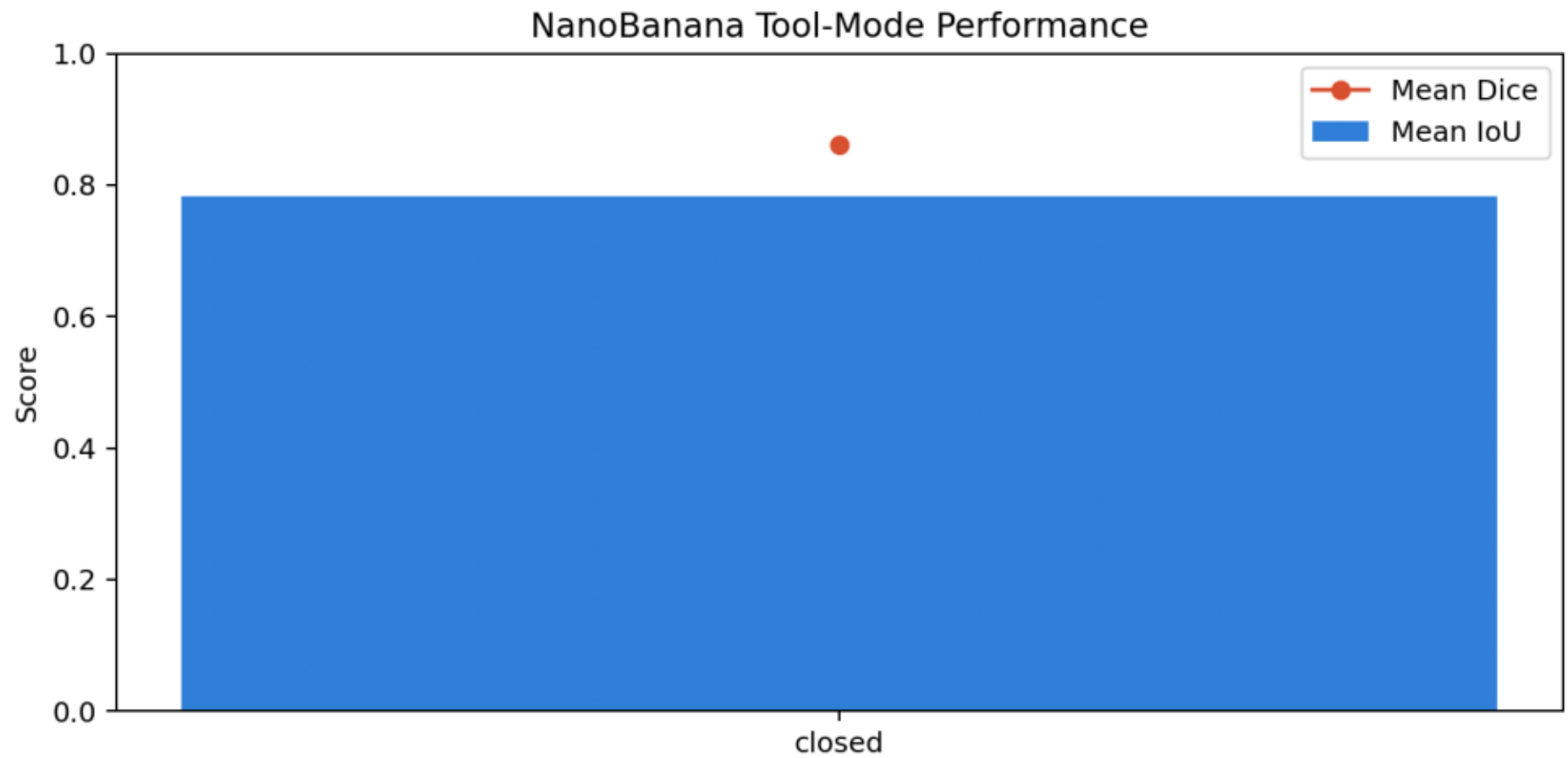


Mean IoU bars with Mean Dice line by tool mode.

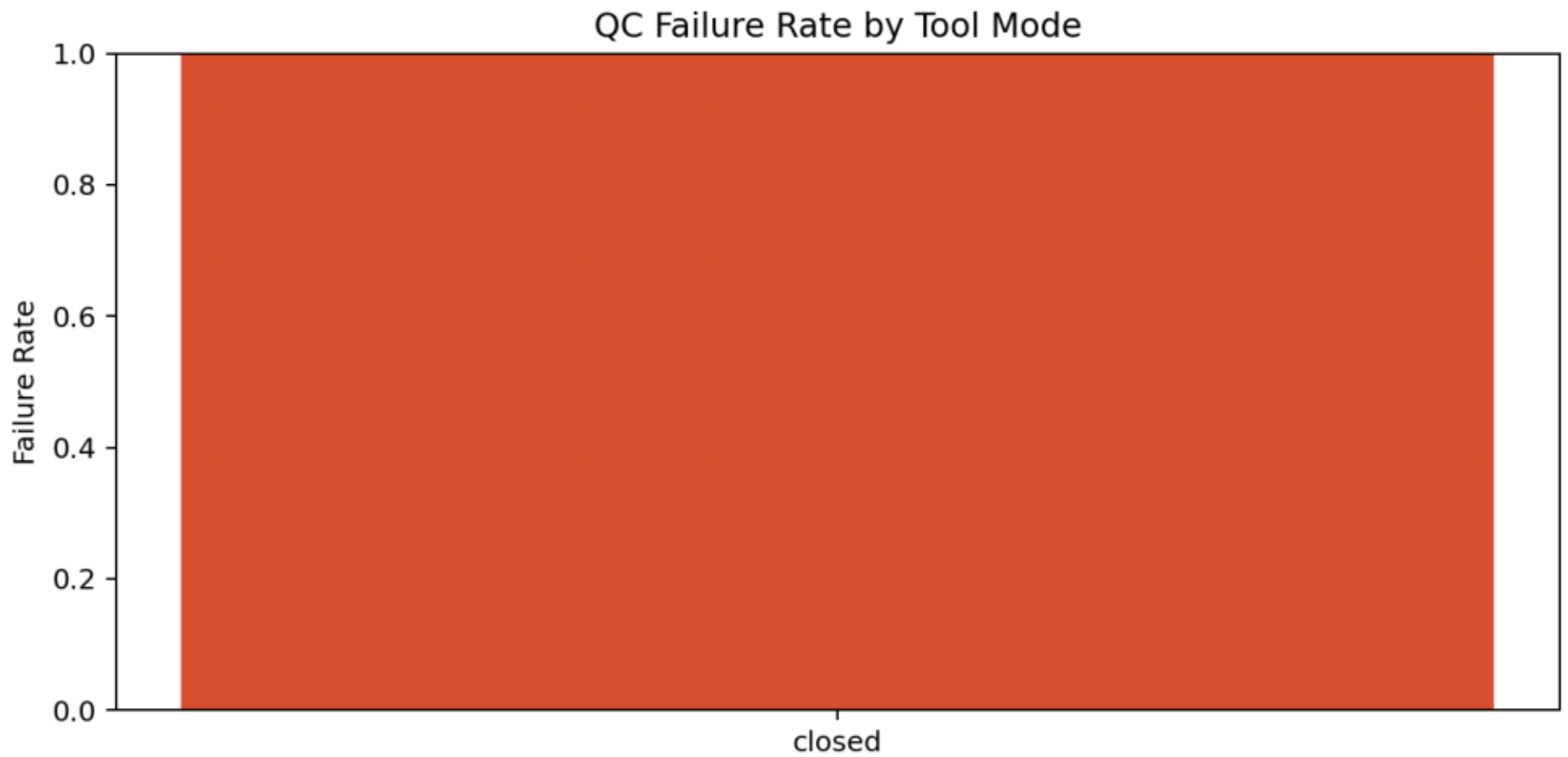
Primary QC Failure Plot



Sensitivity Performance Plot



Sensitivity QC Failure Plot



Sensitivity-set QC failure rate by tool mode.

Per-Run Results (Excerpt)

image_name	tool_mode	iou	dice	precision	recall	qc_pass	selected_atte	retrieval_dup	retrieval_mas	analysis_primar
0004a718-546c	closed	0.3263	0.4921	0.9365	0.3337	0.0000	1.0000	0.0000	0.0000	1.0000
0017b7c7-90f8	closed	0.6696	0.8021	0.9849	0.6765	0.0000	1.0000	0.0000	0.0000	1.0000
0048d8c5-b59c	closed	0.9505	0.9746	0.9951	0.9550	0.0000	1.0000	0.0000	0.0000	1.0000
00f98835-8fd8	closed	0.8604	0.9250	0.9998	0.8606	0.0000	1.0000	0.0000	0.0000	1.0000
01503109-d81f	closed	0.6762	0.8068	0.6938	0.9638	0.0000	1.0000	0.0000	0.0000	1.0000
0198bc50-169e	closed	0.9358	0.9668	0.9889	0.9458	0.0000	1.0000	0.0000	0.0000	1.0000
01c95ee8-79cc	closed	0.9527	0.9758	0.9897	0.9623	0.0000	1.0000	0.0000	0.0000	1.0000
01dbf397-c0fc	closed	0.9546	0.9768	0.9909	0.9630	0.0000	1.0000	0.0000	0.0000	1.0000
01dcc13a-85ab	closed	0.5527	0.7120	0.9923	0.5551	0.0000	1.0000	0.0000	0.0000	1.0000
01e6cfd7-bdb6	closed	0.9576	0.9783	0.9856	0.9711	0.0000	1.0000	0.0000	0.0000	1.0000